EE 121B: Principles of Semiconductor Device Design

Lecture 11:

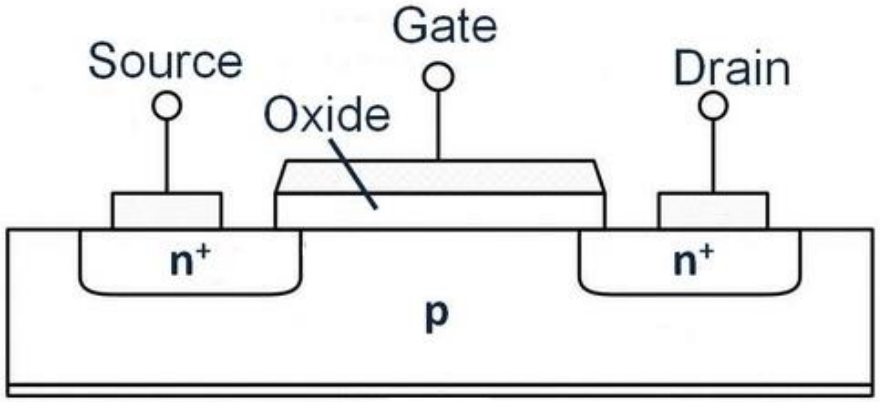
The MOS Capacitor

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Course Overview

1. Semiconductor Properties
2. Metal-Semiconductor Contacts
3. P-N Junction
4. MOS Devices
5. Bipolar Junction Transistor

Metal-Oxide-Semiconductor (MOS)
Field-Effect Transistor (FET)



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Lecture 11

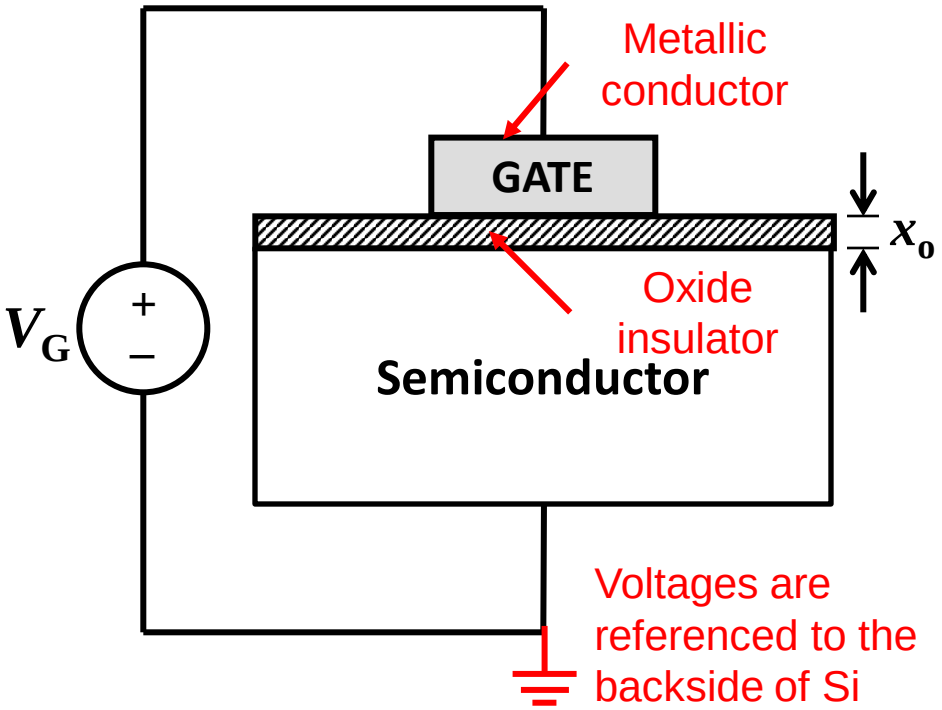
OUTLINE

- The MOS Capacitor
  - Energy band diagrams

Reading: Pierret 16.1-16.2, 18.1; Hu 5.1

MOS Capacitor Structure

MOS capacitor
(cross-sectional view)



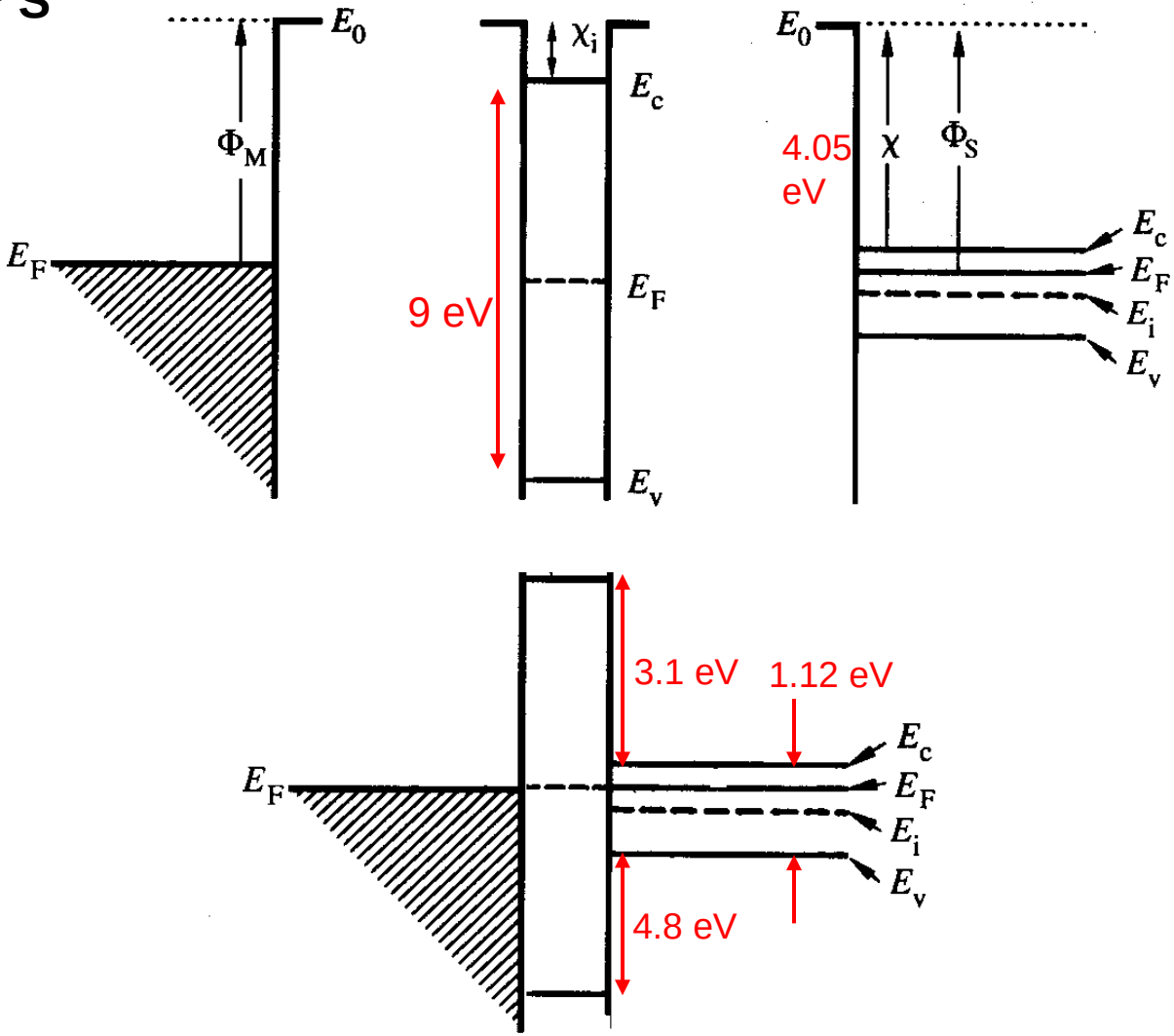
$x_0 < 5\text{nm}$ in modern ICs

- **Most MOS devices today employ:**
  - **degenerately doped polycrystalline Si ("poly-Si") as the "metallic" gate-electrode material**
    - $\Phi_M \sim 4.1$ eV
    - n$^+$-type for "n-channel" transistors
    - p$^+$-type, for "p-channel" transistors
    - $\Phi_M \sim 5.2$ eV
  - **SiO_2 as the gate dielectric**
    - band gap = 9 eV
    - $\varepsilon_{r,SiO2} = 3.9 \approx \frac{1}{3}\varepsilon_{r,Si}$
  - **Si as the semiconductor material**
    - p-type, for "n-channel" transistors
    - n-type, for "p-channel" transistors

Special Case: Equal Work Functions

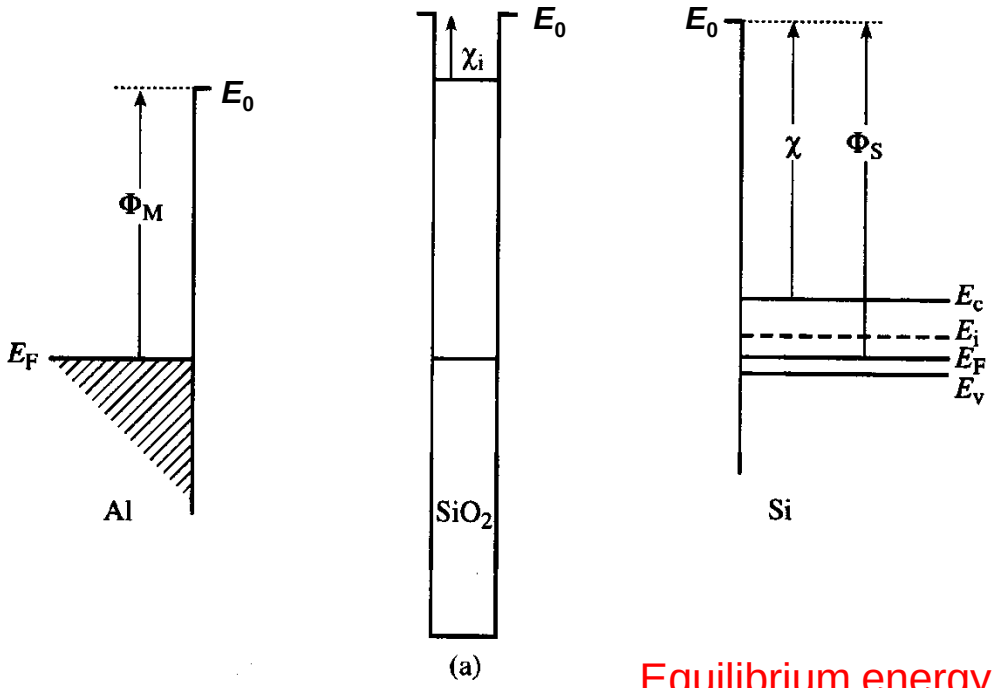
$\Phi_M = \Phi_S$

R. F. Pierret, *Semiconductor Device Fundamentals*, Fig. 16.2



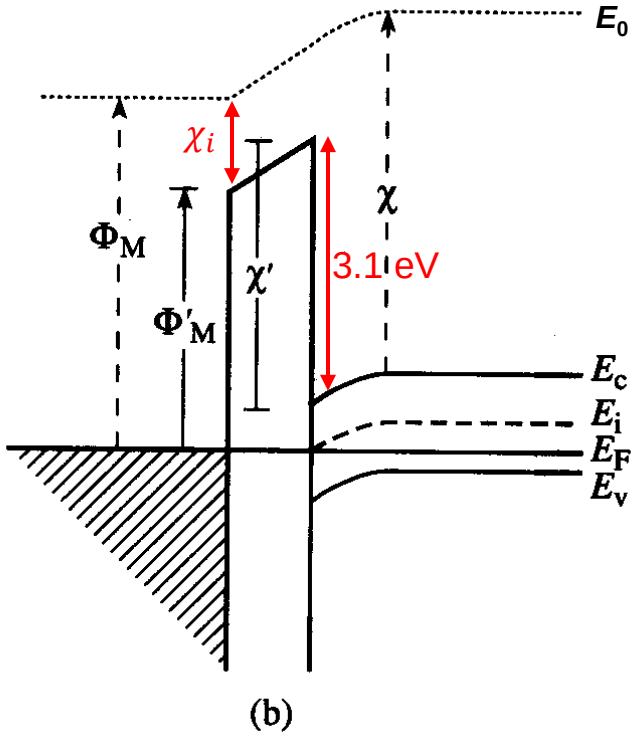
General Case: Different Work Functions

R. F. Pierret, *Semiconductor Device Fundamentals*, Fig. 18.1



- E_0 is a continuous function of distance
- χ is a fundamental material property

Equilibrium energy band diagram:



MOS Band Diagrams: Guidelines

- Fermi level E_F is flat (constant with x) within the semiconductor
  - Since no current flows in the x direction, we can assume that equilibrium conditions prevail

- Band bending is linear within the oxide
  - No charge in the oxide => $d\mathcal{E}/dx = 0$ so $\mathcal{E}$ is constant
    => dE_c/dx is constant

$$\frac{d\mathcal{E}}{dx} = \frac{\rho}{\varepsilon_s}$$

- From Gauss' Law, we know that the electric field strength in the Si at the surface, $\mathcal{E}_{Si}$, is related to the electric field strength in the oxide, $\mathcal{E}_{ox}$: Electric displacement is continuous if there is no charge at the SiO_2/Si interface

$$\varepsilon_{ox}\mathcal{E}_{ox} = \varepsilon_{Si}\mathcal{E}_{Si}$$

$$\rightarrow \mathcal{E}_{ox} = \frac{\varepsilon_{Si}}{\varepsilon_{ox}}\,\mathcal{E}_{Si} \cong 3\,\mathcal{E}_{Si} \qquad so \quad \left.\frac{dE_c}{dx}\right|_{oxide} = 3 \times \left.\frac{dE_c}{dx}\right|_{Si\,(at\ the\ surface)}$$

MOS Band Diagram Guidelines (cont'd)

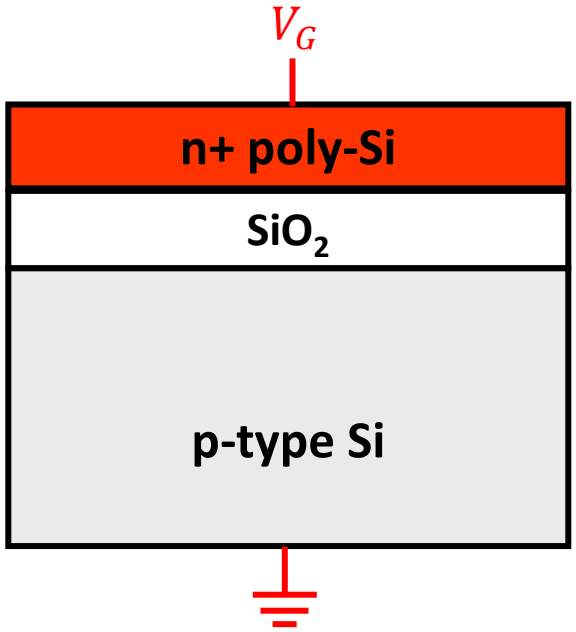
- The barrier height for conduction-band electron flow from the Si into SiO_2 is 3.1 eV
  - This is equal to the electron-affinity difference (χ_{Si} and χ_{SiO2})

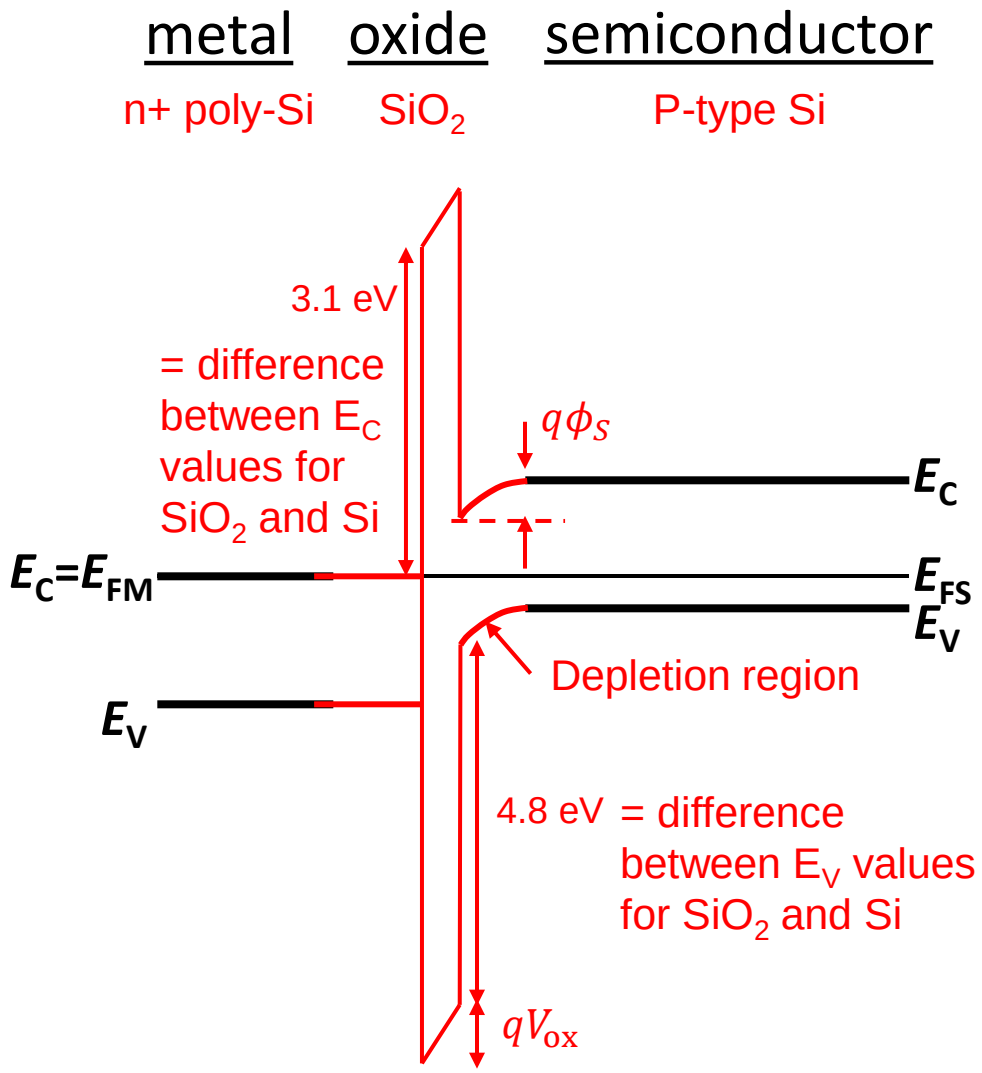
- The barrier height for valence-band hole flow from the Si into SiO_2 is 4.8 eV

- The vertical distance between the Fermi level in the metal, E_{FM}, and the Fermi level in the Si, E_{FS}, is equal to the applied gate voltage (assuming that the Si bulk is grounded):

$$qV_G = E_{FS} - E_{FM}$$

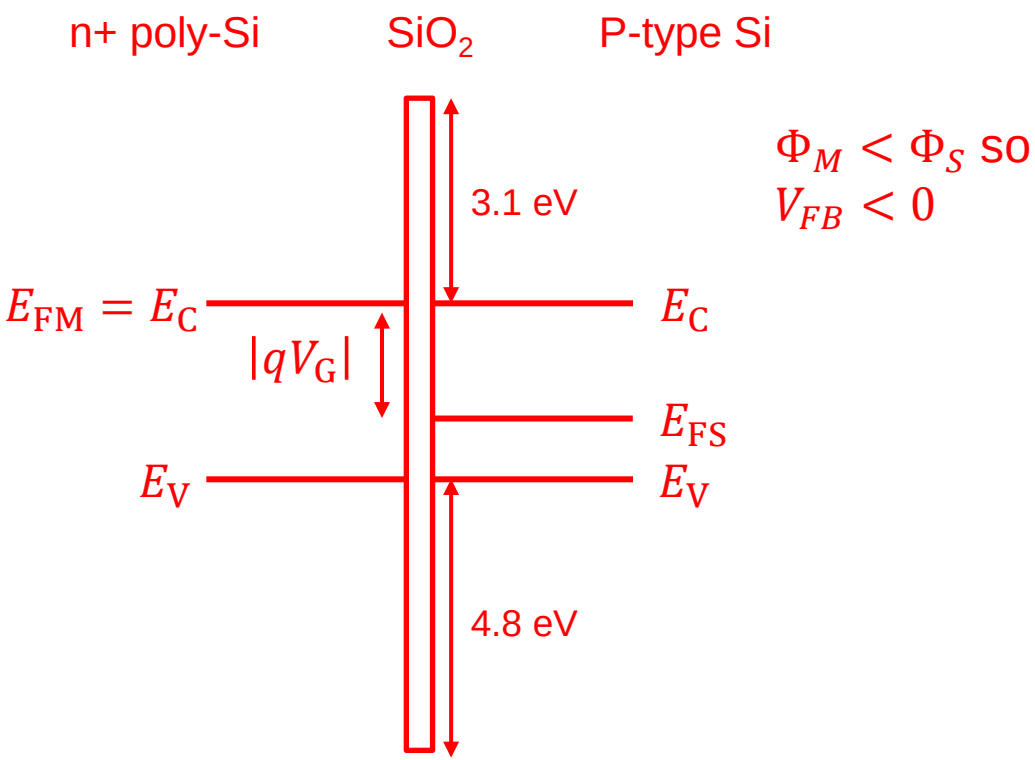
MOS Equilibrium Band Diagram





Flat-Band Condition

- The **flat-band voltage, V_{FB},** is the applied voltage which results in no band-bending within the semiconductor.
  - Ideally, this is equal to the work-function difference between the gate and the bulk of the semiconductor: $qV_{FB} = \Phi_M - \Phi_S$



Voltage Drops in the MOS System

- In general,

Voltage dropped within oxide

Voltage dropped within Si

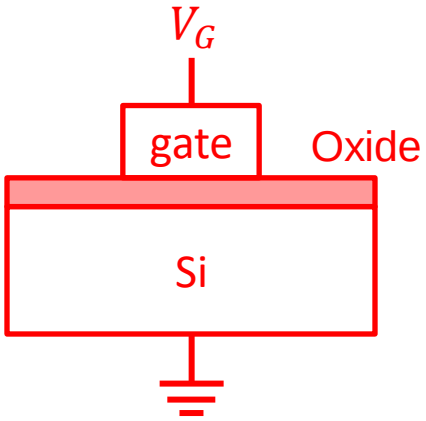
$$V_G = V_{FB} + V_{ox} + \phi_s$$

Flat band voltage
(analogous to built-in potential)

where

$$qV_{FB} = \Phi_{MS} = \Phi_M - \Phi_S$$

V_{ox} is the voltage dropped across the oxide
(V_{ox} = total amount of band bending in the oxide)



ϕ_s is the voltage dropped in the silicon
(total amount of band bending in the silicon)

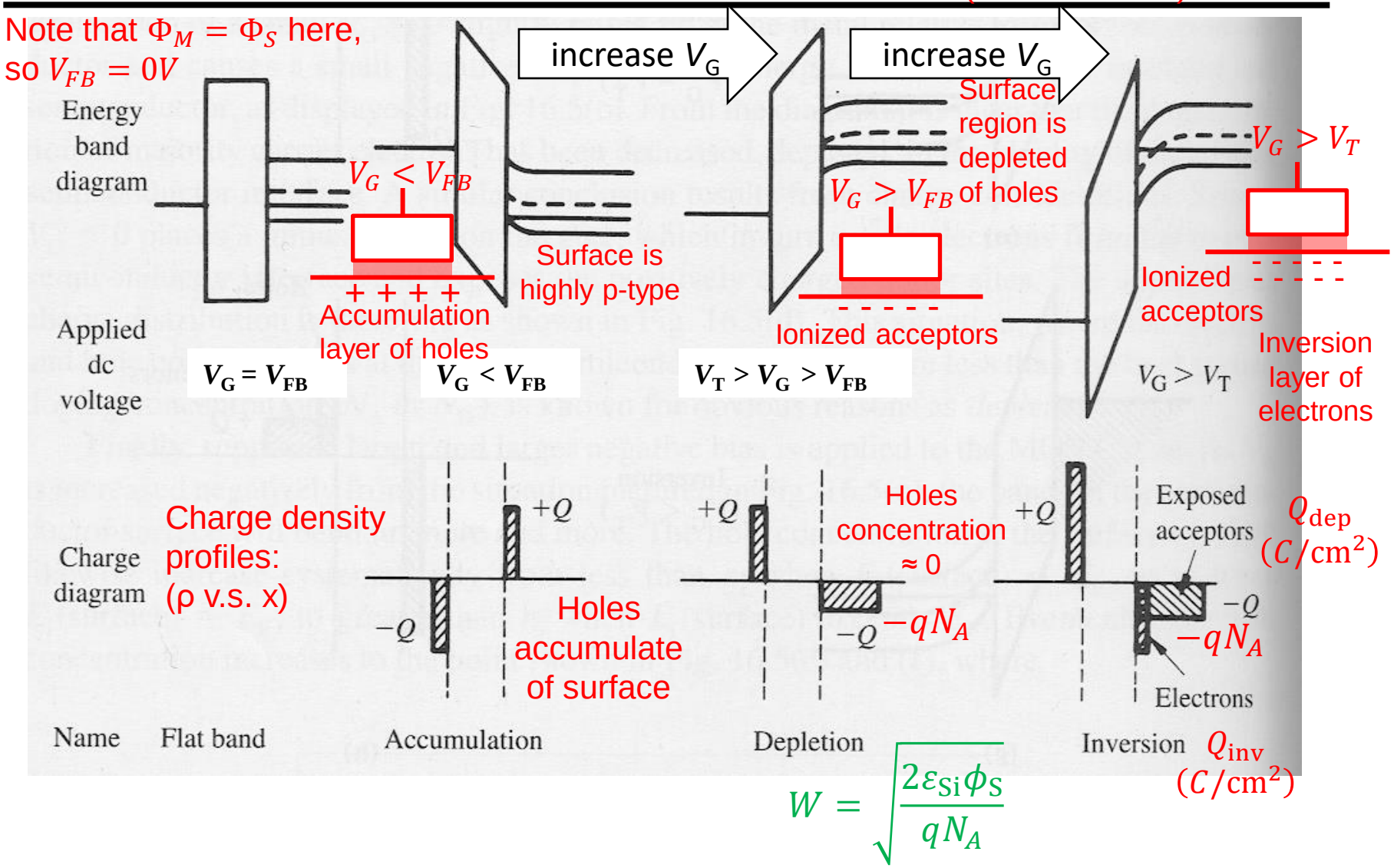
$$q\phi_S = E_i(bulk) - E_i(surface)$$

- For example:

When $V_G = V_{FB}$, $V_{ox} = \phi_s = 0$, *i.e.* there is no band bending

MOS Operating Regions (p-type Si)

(NMOS device)



$$W = \sqrt{\frac{2\varepsilon_{Si}\phi_S}{qN_A}}$$

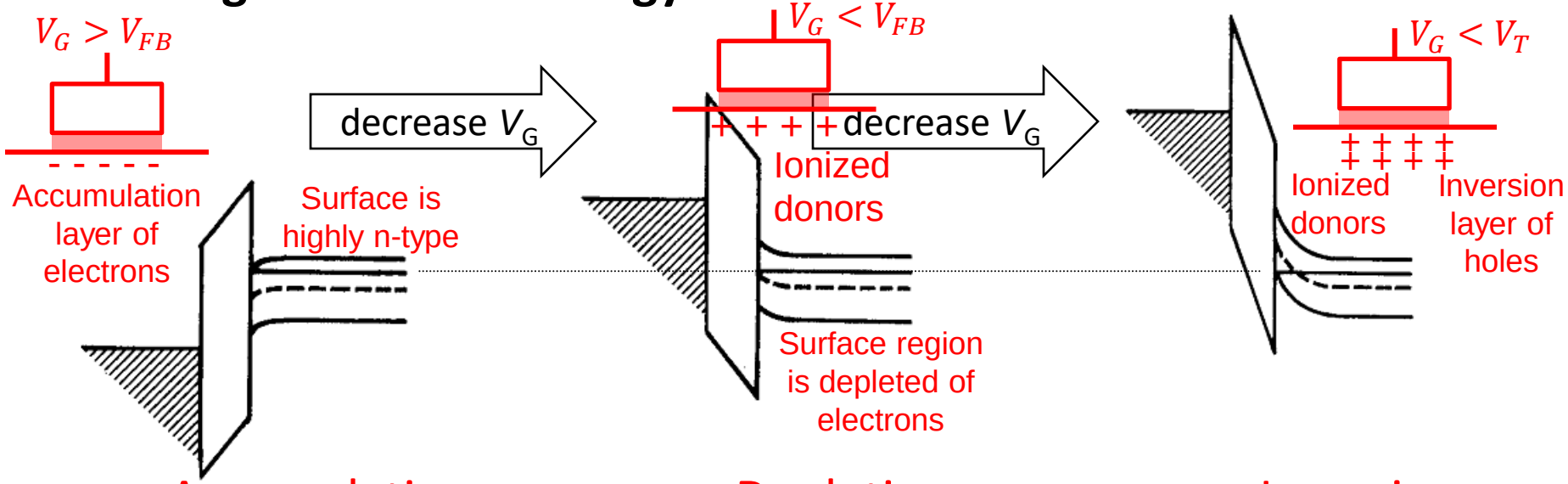
R. F. Pierret, *Semiconductor Device Fundamentals*, Fig. 16.6

MOS Operating Regions (n-type Si)

(PMOS device)

Decrease V_G toward more negative values

→ the gate electron energy increases relative to that in the Si



- **Accumulation**

  $V_G > V_{FB}$

  Electrons accumulated at Si surface

- **Depletion**

  $V_G < V_{FB}$

  Electrons depleted from Si surface

- **Inversion**

  $V_G < V_T$

  Surface inverted to p-type

R. F. Pierret, *Semiconductor Device Fundamentals*, Fig. 16.5

Acknowledgement

- Modern Semiconductor Devices for Integrated Circuits by C. Hu (Prentice Hall, 2009)
- Semiconductor Device Fundamentals by R. F. Pierret (Addison Wesley, 1996)
- Professor Tsu-Jae King Liu EE130/230A notes